

Distributed Systems Concerns in Practice

Fabrics and topology:

- Intra-node: PCIe, NVLink
- Inter-node: 25/100/200/400 Gbps Ethernet/InfiniBand; latencies 2–10× higher

Collectives vs parameter server:

- Collectives (NCCL/Gloo/MPI): bandwidth-efficient, symmetric
- PS: flexible, supports async, but can bottleneck at servers

Orchestration & launch:

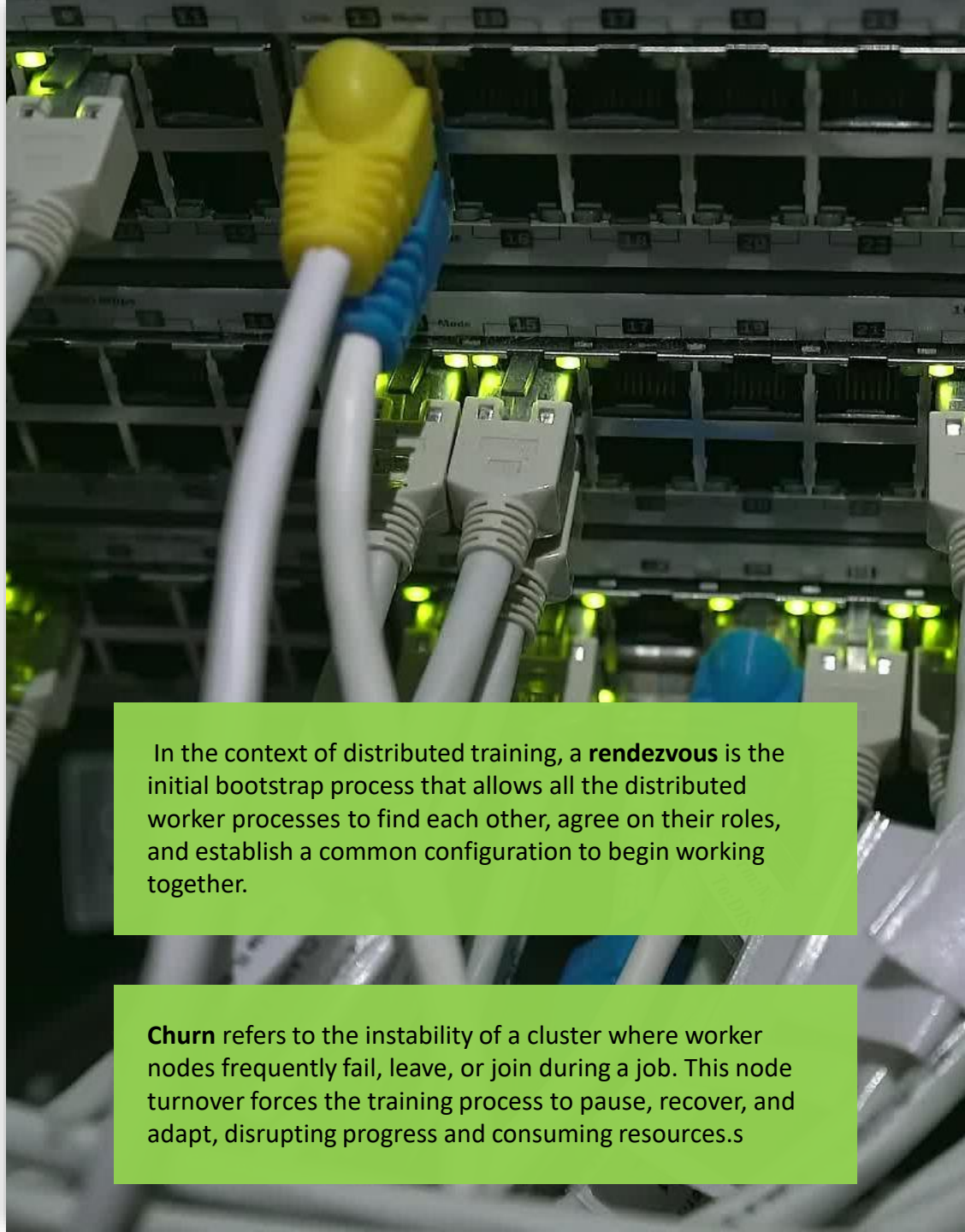
- SLURM/Kubernetes; ranks, world size, rendezvous; topology-aware placement

Reproducibility:

- Deterministic sharding and seeding; avoid nondeterministic kernels

Elasticity:

- Handle node churn; reshard data and adjust LR safely



In the context of distributed training, a **rendezvous** is the initial bootstrap process that allows all the distributed worker processes to find each other, agree on their roles, and establish a common configuration to begin working together.

Churn refers to the instability of a cluster where worker nodes frequently fail, leave, or join during a job. This node turnover forces the training process to pause, recover, and adapt, disrupting progress and consuming resources.

Sync vs Async vs Elastic Data Parallel

Synchronous all-reduce:

- Pros: stable convergence; single-node equivalence
- Cons: straggler-sensitive; per-step barrier

Parameter-server (async/stale-sync):

- Pros: more throughput; tolerant to slow clients
- Cons: stale gradients; careful LR tuning; potential accuracy hit

Elastic DP (synchronous with changing world size):

- World size adjusts at runtime; reshard data/checkpoints; adjust LR with effective batch

Guidance: prefer sync for accuracy-critical; use async/elastic when churn is inevitable

Topology-Aware & Hierarchical All-Reduce; Libraries

- Libraries:
 - NCCL (GPU-centric), Gloo (CPU/GPU), MPI (HPC)
- Algorithms:
 - Ring: bandwidth-optimal for large messages
 - Tree: lower latency for smaller messages
 - Reduce-scatter + all-gather: core of sharded training
- Hierarchical all-reduce:
 - Intra-node reduce-scatter (NVLink)
 - Inter-node reduce-scatter (IB/Ethernet)
 - Inter-node all-gather
 - Intra-node all-gather
- Practice: bucket sizes, algorithm selection per tensor size, topology hints